DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

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AIR QUALITY ANALYSIS

Submitted by

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CERTIFICATE

This is to certify that THAHSEEN JAFAR PP bearing Registration no. 12414070 has completed INT375 project titled, **"Air Quality Analysis"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

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DECLARATION

I, THAHSEEN JAFAR PP student of BTECH CSE under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20-04-2026 Signature

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INTRODUCTION

Air pollution has emerged as one of the most critical environmental challenges in recent decades, particularly in rapidly developing countries like India. The increasing levels of industrialization, urbanization, vehicular emissions, and population growth have significantly contributed to the deterioration of air quality across major cities and states. Poor air quality not only affects environmental sustainability but also poses severe risks to human health, leading to respiratory diseases, cardiovascular problems, and reduced life expectancy. Therefore, monitoring and analysing air pollution has become essential for policymakers, researchers, and public health authorities.

In India, air quality varies significantly across regions due to differences in geography, climate conditions, industrial activities, and urban density. Metropolitan regions such as Delhi are often reported among the most polluted areas due to high vehicular density, construction activities, and unfavourable meteorological conditions that trap pollutants. On the other hand, states like Kerala, with more greenery and coastal influence, generally exhibit comparatively better air quality. Maharashtra presents a mixed scenario, with cities like Mumbai experiencing moderate to high pollution levels due to industrial and urban activities. This variation makes it important to conduct a comparative study across different states to understand patterns and trends in air pollution.

This project focuses on the comparative time series analysis of air quality in three Indian states: Kerala, Delhi, and Maharashtra, over the period from 2010 to 2023. The dataset used in this study consists of multiple air pollutant parameters such as PM2.5, PM10, nitrogen oxides (NO, NO_2, NOx), sulfur dioxide (SO_2), carbon monoxide (CO), ozone, ammonia (NH_3), and benzene. Among these, PM2.5 is considered a key indicator of air pollution due to its ability to penetrate deep into the lungs and bloodstream, making it highly harmful to human health. In the absence of a direct Air Quality Index (AQI) in the dataset, PM2.5 levels are used as the primary metric for analysis.

The project involves collecting data from multiple sources, merging and pre-processing it using Python libraries such as Pandas and NumPy, and performing exploratory data analysis (EDA) using visualization tools like Matplotlib and Seaborn. Various analytical techniques are applied to identify trends, seasonal patterns, correlations between pollutants, and the presence of outliers. Time-based features such as year and month are extracted to study long-term and seasonal variations in air quality.

The primary objective of this analysis is to gain meaningful insights into how air pollution levels have changed over time and how they differ across selected states. By comparing pollution trends and identifying key influencing factors, the study aims to provide a better understanding of environmental conditions in different regions of India. The findings of this project can be useful for raising awareness, supporting policy decisions, and encouraging further research in environmental data analysis.

Overall, this project demonstrates the practical application of data science techniques in analysing real-world environmental data, highlighting the importance of data-driven approaches in addressing critical societal issues such as air pollution.

OBJECTIVES

The primary aim of this project is to analyze and compare air quality across selected Indian states using data science techniques. The study focuses on identifying patterns, trends, and relationships between various air pollutants over time. The specific objectives of the project are as follows:

1. State-wise Air Quality Comparison

To compare the air quality levels across the selected states—Kerala, Delhi, and Maharashtra—by analysing pollutant concentrations. This objective aims to identify which state has relatively higher or lower pollution levels and understand regional variations in air quality. This comparison provides a clear overview of geographical differences in pollution and highlights areas requiring more environmental attention.

2. Year-wise Pollution Trend Analysis

To examine how air pollution levels have changed over the years from 2010 to 2023. This includes identifying increasing or decreasing trends in pollutant concentrations, helping to understand long-term environmental changes and the effectiveness of pollution control measures. Analysing these trends helps in evaluating the impact of policies and identifying whether air quality is improving or deteriorating over time.

3. Seasonal (Monthly) Pattern Analysis

To analyse the variation in air pollution levels across different months of the year. This objective helps in identifying seasonal patterns, such as higher pollution during winter months or lower levels during monsoon seasons, and understanding the influence of climatic factors on air quality. Understanding seasonal patterns is useful for predicting pollution levels and planning preventive measures during high-risk periods.

4. Correlation between Pollutants

To study the relationship between different air pollutants such as PM2.5, PM10, NO_2, SO_2, and others. This involves using correlation analysis to determine how pollutants are interrelated and whether an increase in one pollutant is associated with increases in others. This analysis helps in identifying key pollutants that influence overall air quality and supports better environmental monitoring strategies.

5. Outlier Detection in Pollution Levels

To identify unusual or extreme values in the dataset that deviate significantly from normal pollution levels. Detecting outliers helps in understanding abnormal environmental events, data inconsistencies, or sudden spikes in pollution that may require further investigation. Recognizing such anomalies is important for ensuring data reliability and for highlighting critical pollution events that may need immediate attention.

SOURCE OF DATASET

The dataset used in this project was obtained from Kaggle, a well-known platform for data science that provides access to a wide range of structured datasets for analysis and research. The selected dataset contains historical air quality data for various cities across India covering the period from 2010 to 2023. This dataset is originally sourced from the Central Pollution Control Board (CPCB), which operates under the Ministry of Environment, Forest and Climate Change, Government of India.

The CPCB is responsible for monitoring air pollution levels across the country through a network of Continuous Ambient Air Quality Monitoring Stations (CAAQMS). These stations collect data on multiple air pollutants at regular intervals, ensuring that the information is reliable and suitable for analytical purposes. The dataset available on Kaggle is a compiled and cleaned version of this raw CPCB data, making it easier to use for data analysis.

The dataset includes key air quality parameters such as PM2.5, PM10, nitrogen oxides (NO, NO_2, NOx), sulfur dioxide (SO_2), carbon monoxide (CO), ozone, ammonia (NH_3), and benzene. These pollutants are commonly used to assess air quality and understand environmental conditions. For this project, data corresponding to three states—Kerala, Delhi, and Maharashtra—was selected in order to perform a comparative analysis across different regions.

The dataset was initially provided in multiple CSV files categorized by monitoring stations. These files were combined into a single dataset using Python for further processing. Since the dataset does not contain a direct Air Quality Index (AQI), PM2.5 concentration was used as the primary indicator of air quality due to its significant impact on human health.

Overall, the dataset is considered reliable and appropriate for analysis, as it is derived from official CPCB records and made accessible through Kaggle in a structured format.

ETL PROCESS

The ETL (Extract, Transform, Load) process is an essential step in data analysis, as it involves collecting data from the source, cleaning and preparing it for analysis, and finally storing it in a usable format. In this project, the ETL process was performed using Python libraries such as Pandas and NumPy to handle and pre-process the air quality dataset efficiently.

1. Data Extraction

The data for this project was extracted from Kaggle, where the dataset was available in the form of multiple CSV files. These files contained air quality data collected from different monitoring stations across various states in India. Each file represented a specific station or region and included pollutant measurements over a period of time.

The dataset was downloaded as a compressed ZIP file and then extracted into a local directory. Using Python, all CSV files corresponding to the selected states—Kerala, Delhi, and Maharashtra—were accessed from the folder. The data was read using the Pandas library, which allowed efficient handling of large datasets. Only the required columns, such as pollutant concentrations and date fields, were selected for further analysis.

2. Data Transformation

After extraction, the dataset underwent several pre-processing and transformation steps to ensure it was clean and suitable for analysis. Initially, column names were standardized by removing extra spaces and renaming them into a consistent format for easier handling. Since the dataset was spread across multiple files, all files were merged into a single DataFrame to create a unified dataset.

Missing values were identified and handled by replacing them with the mean of the respective columns, ensuring that the dataset remained consistent without losing important information. Duplicate records were also removed to avoid redundancy. The date column was converted into a proper date-time format, and additional features such as year and month were extracted to support time-based analysis.

Furthermore, unnecessary columns were removed, and relevant pollutant variables such as PM2.5, PM10, NO_2, SO_2, and others were retained. Data types were optimized by converting categorical columns like state and station identifiers into category types, which improved memory efficiency and processing speed.

3. Data Loading

After completing the transformation process, the cleaned and processed dataset was stored in a structured format for further analysis. The combined dataset was saved as a CSV file, allowing it to be easily reloaded without repeating the extraction and transformation steps.

Additionally, a sampled version of the dataset was created to improve computational efficiency during exploratory data analysis. This step ensured faster processing while still maintaining the overall data distribution and trends. The final dataset was then used for visualization and analysis using tools such as Matplotlib and Seaborn.

ANALYSIS

The analysis phase of this project focuses on exploring and interpreting the air quality data for the selected states—Kerala, Delhi, and Maharashtra—using various data analysis and visualization techniques. After completing the ETL process, the cleaned dataset was examined to identify patterns, trends, and relationships among different air pollutants. Python libraries such as Pandas, Matplotlib, and Seaborn were utilized to perform exploratory data analysis (EDA) and generate meaningful visualizations that aid in better understanding the data.

Since the dataset does not include a direct Air Quality Index (AQI), PM2.5 concentration was used as the primary indicator of air quality. PM2.5 is widely considered a critical pollutant due to its harmful impact on human health, as it can penetrate deep into the lungs and bloodstream. Therefore, analysing PM2.5 levels provides a reliable representation of overall air pollution conditions.

The analysis was carried out from multiple perspectives to gain comprehensive insights. Initially, the distribution of PM2.5 values was studied to understand the spread and variability of pollution levels. This helped in identifying whether the data is normally distributed or skewed, as well as detecting the presence of extreme values. Next, a comparative analysis across states was performed to evaluate differences in pollution levels between Kerala, Delhi, and Maharashtra. This comparison highlighted regional variations and helped identify areas with relatively higher pollution.

Time-based analysis was also conducted by examining year-wise trends to understand how pollution levels have changed over the period from 2010 to 2023. This provided insights into long-term environmental patterns and the effectiveness of pollution control measures. Additionally, monthly analysis was performed to identify seasonal variations in air quality, revealing how pollution levels fluctuate throughout the year due to climatic factors.

Furthermore, correlation analysis was used to study the relationships between different pollutants, helping to understand how they influence each other. A scatter plot analysis between PM2.5 and PM10 was also conducted to examine their relationship in more detail. Overall, this comprehensive analysis enables a deeper understanding of air pollution patterns and supports data-driven conclusions regarding environmental conditions in the selected states.

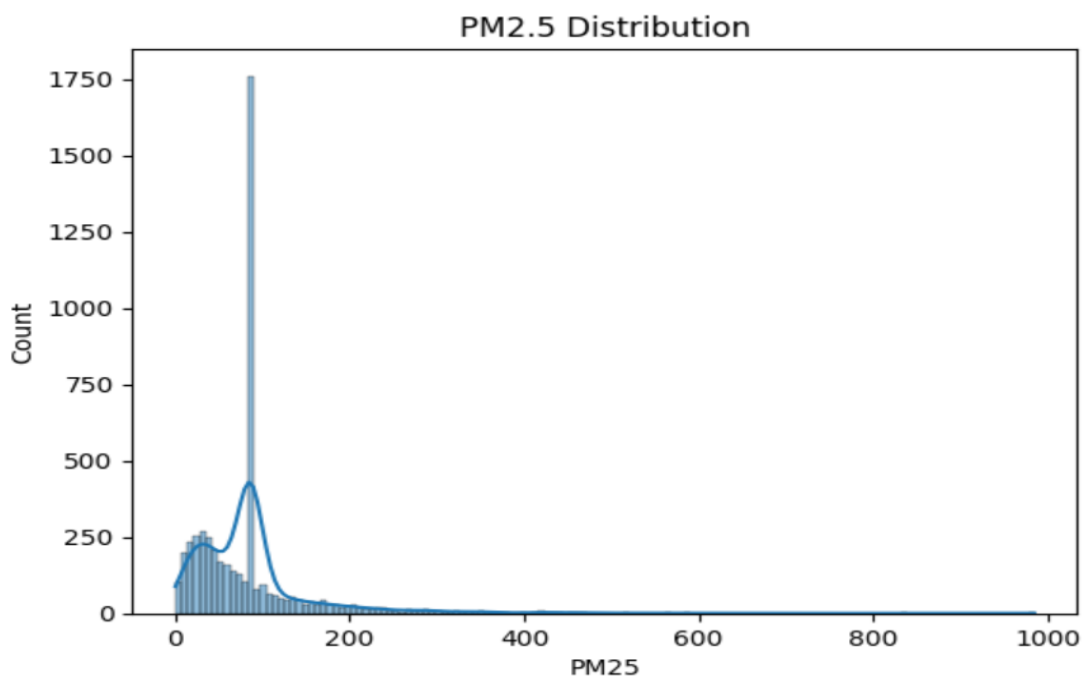
LIST OF ANALYSIS WITH RESULTS

1. Distribution of PM2.5 Levels

The distribution of PM2.5 levels was analysed using a histogram combined with a kernel density estimation (KDE) curve to understand how pollution values are spread across the dataset. The visualization indicates that most PM2.5 values are concentrated within a moderate range, while fewer observations extend toward higher values. This suggests a positively skewed distribution, where extreme pollution levels occur less frequently but are still significant.

The presence of higher-end values indicates occasional spikes in pollution, which may be caused by factors such as increased traffic, industrial emissions, or seasonal effects like winter smog. The KDE curve further supports this pattern by showing a peak at lower values and a gradual decline toward higher concentrations.

This analysis helps in understanding the overall behaviour of PM2.5 levels and identifying the presence of outliers. It provides a foundation for further analysis by highlighting how pollution levels vary across the dataset.

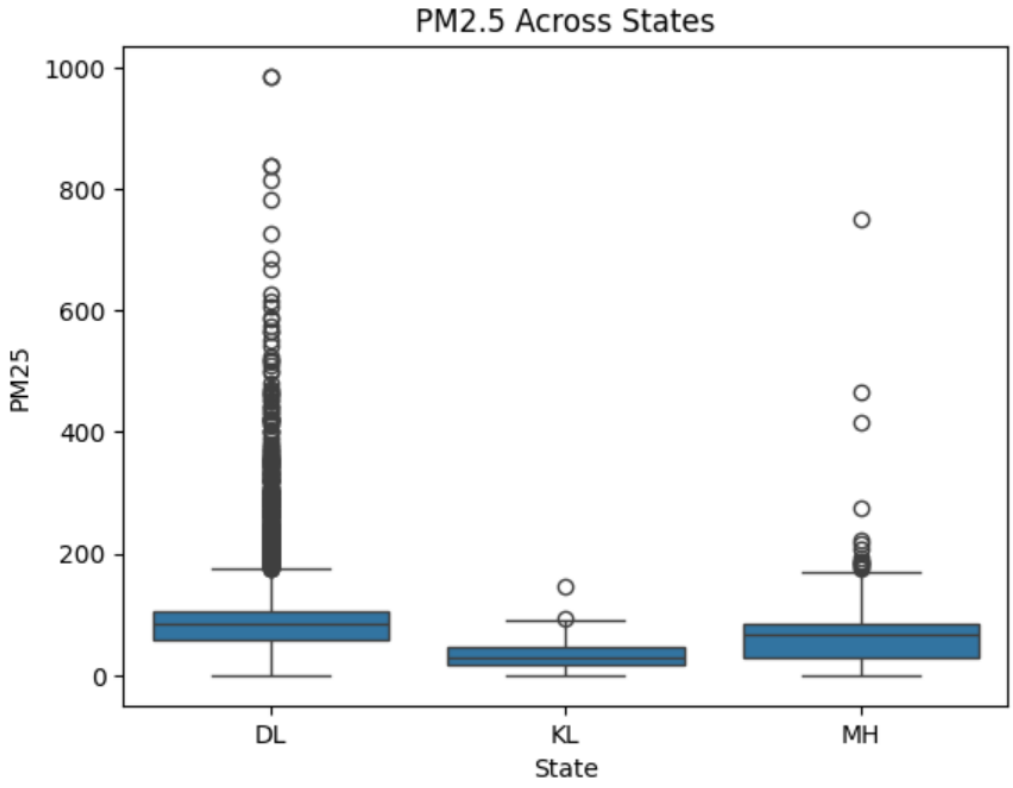


2. State-wise Comparison of Pollution

The state-wise comparison of PM2.5 levels was performed using a boxplot to visualize the distribution, median, and variability across Kerala, Delhi, and Maharashtra. The results show that Delhi has the highest median PM2.5 levels along with a wider range, indicating consistently higher pollution and greater variation. Kerala, in contrast, exhibits lower pollution levels with a more compact distribution, suggesting better air quality.

Maharashtra lies between the two, showing moderate pollution levels with some variability. The presence of outliers in the plot indicates occasional extreme pollution events in all states, especially in Delhi. These differences can be attributed to variations in population density, industrial activity, and environmental conditions.

This analysis clearly highlights regional differences in air quality and helps identify areas that require more focused pollution control measures.

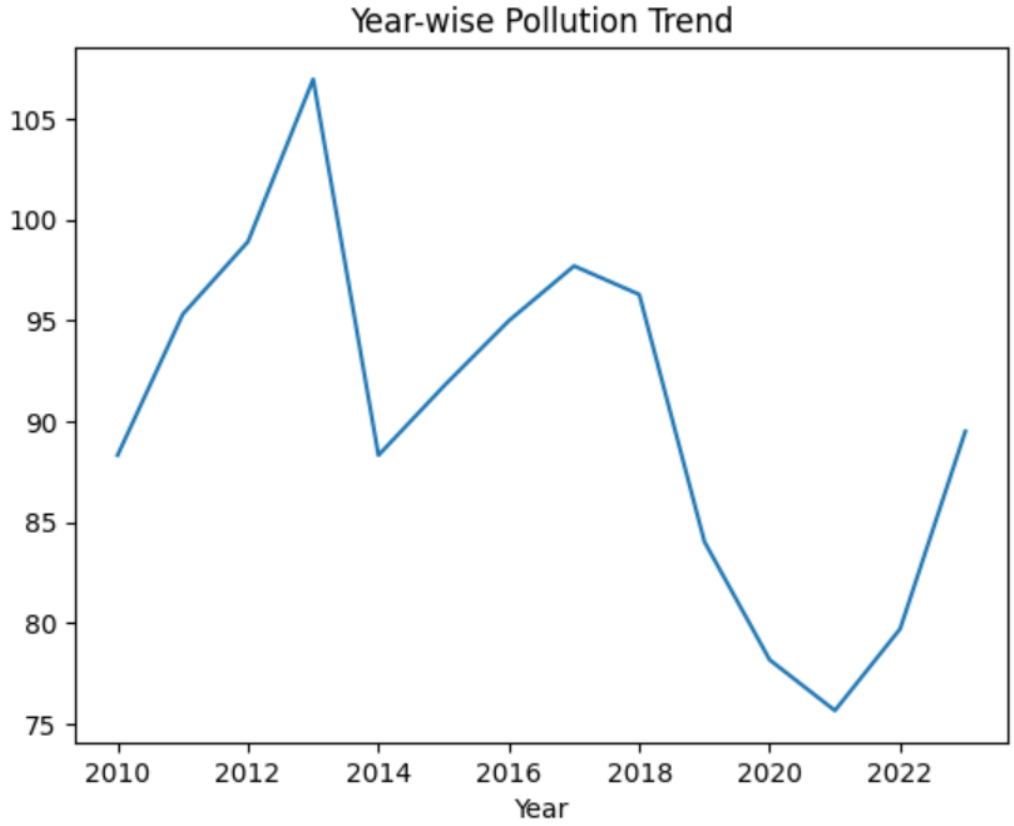


3. Year-wise Trend Analysis

The year-wise trend analysis was conducted using a line plot showing the average PM2.5 levels from 2010 to 2023. The results reveal noticeable fluctuations in pollution levels over the years, with certain periods showing increases and others showing slight declines. These variations may be influenced by factors such as policy changes, urban development, and industrial growth.

In some years, pollution levels appear to rise, indicating increased environmental pressure, while in other years, slight improvements can be observed, possibly due to government initiatives and awareness programs. However, the trend does not show a consistent decline, suggesting that air pollution remains a persistent issue.

This analysis is useful for understanding long-term changes in air quality and evaluating the effectiveness of pollution control measures over time.

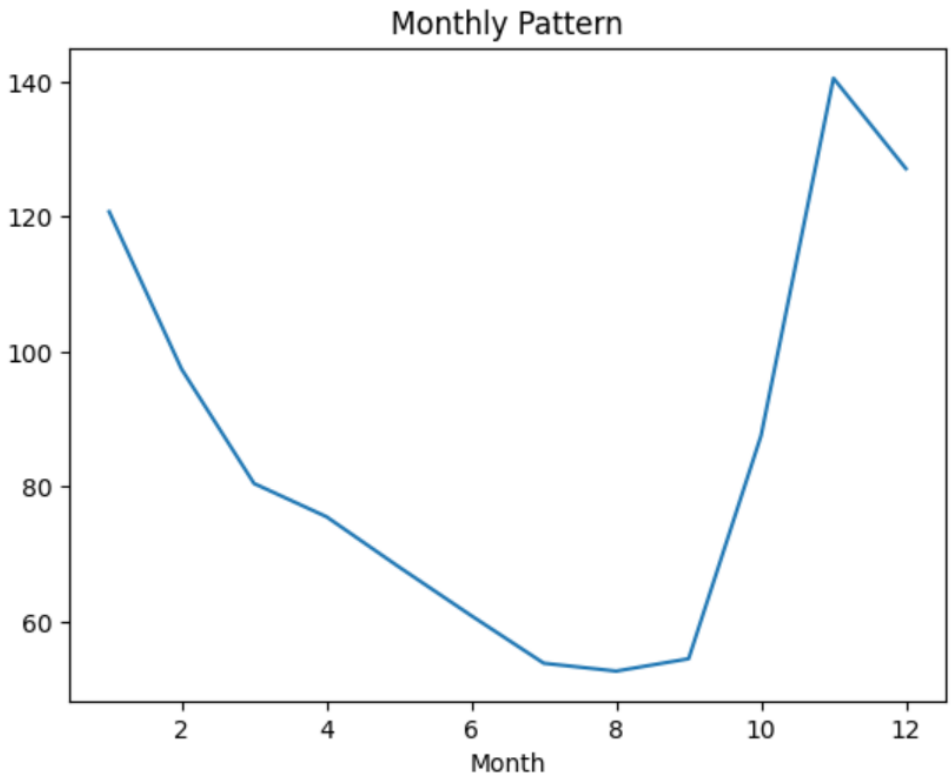


4. Monthly Pattern Analysis

The monthly pattern analysis was performed to study seasonal variations in PM2.5 levels using a line plot. The results indicate that pollution levels vary across different months, showing a clear seasonal pattern. Higher pollution levels are generally observed during winter months, which can be attributed to lower temperatures, reduced wind speed, and conditions that trap pollutants near the ground.

In contrast, lower pollution levels are often seen during monsoon months due to rainfall, which helps remove pollutants from the atmosphere. These variations highlight the influence of climatic factors on air quality.

Understanding these seasonal trends is important for predicting high pollution periods and implementing preventive measures. This analysis helps in identifying critical months where pollution control efforts need to be strengthened.

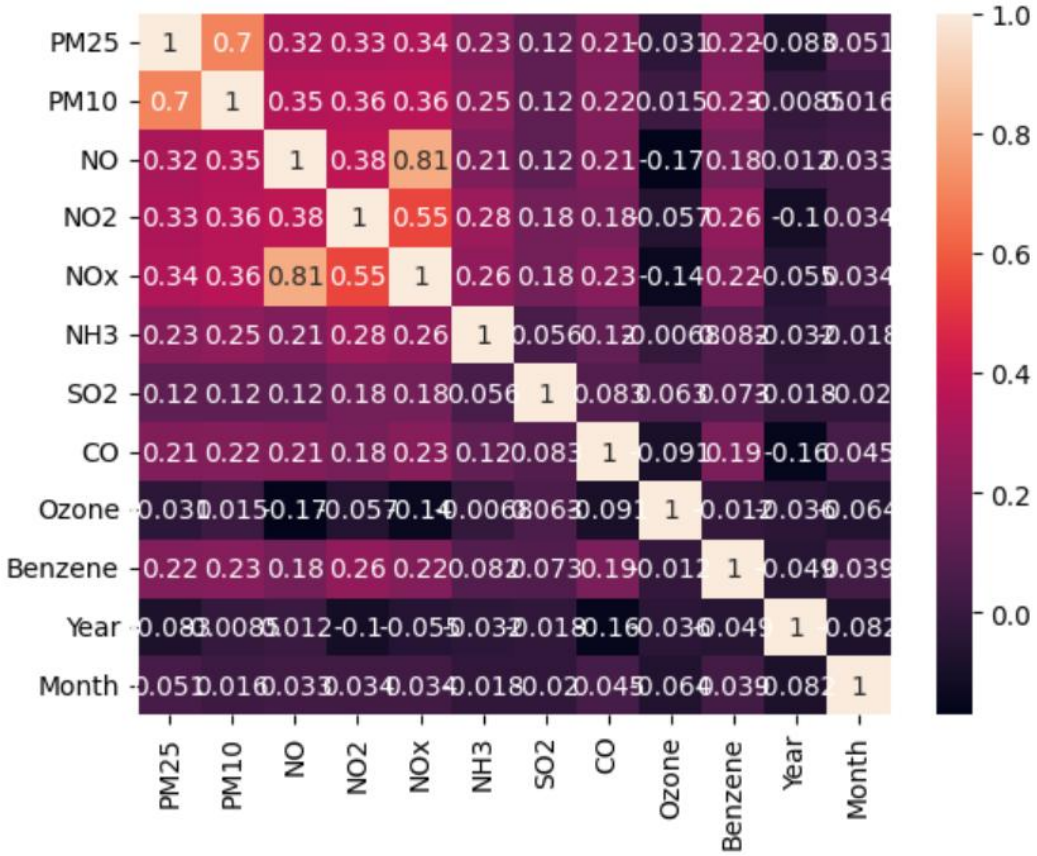


5. Correlation Analysis of Pollutants

Correlation analysis was conducted using a heatmap to examine the relationships between different pollutants in the dataset. The results show that certain pollutants, particularly PM2.5 and PM10, have a strong positive correlation, meaning they tend to increase or decrease together. This suggests that they may originate from similar sources such as vehicle emissions and industrial activities.

Other pollutants, such as NO_2 and SO_2, show moderate correlations, indicating partial relationships influenced by combustion processes. Some pollutants may show weak correlations, suggesting independent behaviour.

This analysis helps in understanding how pollutants are interrelated and which ones have a stronger influence on overall air quality. It is useful for identifying key pollutants that should be targeted for effective pollution control.

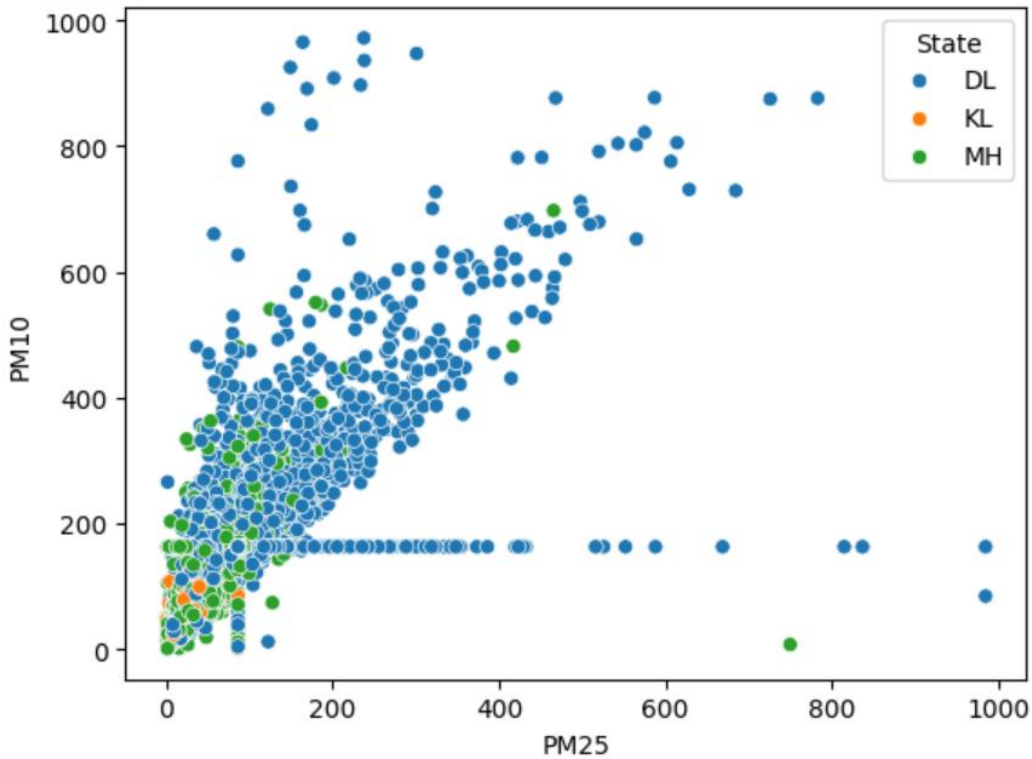


6. Scatter Analysis between PM2.5 and PM10

The relationship between PM2.5 and PM10 was analysed using a scatter plot, which shows how these two variables vary together. The plot reveals a clear positive relationship, where higher PM2.5 values are generally associated with higher PM10 values. This indicates that both pollutants tend to increase simultaneously.

The clustering of points along an upward trend suggests a strong association between the two variables. This is expected, as both PM2.5 and PM10 are particulate matter pollutants that often originate from similar sources such as dust, vehicle emissions, and industrial activities.

This analysis supports the findings from the correlation study and provides a visual understanding of the relationship between the two pollutants. It highlights the importance of controlling particulate matter to improve overall air quality.



FUTURE SCOPE

The present study focuses on analysing air quality trends across selected Indian states using historical data and exploratory data analysis techniques. While the project provides valuable insights into pollution patterns, there is considerable scope for further enhancement and expansion. One of the primary areas for future work is the inclusion of a larger number of states and cities across India. Extending the analysis to a nationwide level would provide a more comprehensive understanding of regional variations in air pollution and help identify areas that require immediate attention.

Another important direction for future research is the integration of real-time air quality data using application programming interfaces (APIs). By incorporating live data streams, the project can be transformed into a dynamic system capable of monitoring current pollution levels. This would allow the development of real-time dashboards and alert mechanisms that can notify users about hazardous air conditions, making the project more practical and impactful.

In addition, the application of advanced machine learning and predictive modelling techniques can significantly enhance the analytical capabilities of the project. Models such as regression analysis, time series forecasting, and deep learning algorithms can be used to predict future air quality levels based on historical data. This would enable early warning systems and assist policymakers in taking preventive measures to control pollution.

The project can also be improved by incorporating additional variables such as weather conditions (temperature, humidity, wind speed), traffic density, and industrial activity. These factors play a crucial role in influencing air quality and can provide deeper insights into the causes of pollution. Furthermore, integrating geographic information systems (GIS) and map-based visualizations can help in spatial analysis, allowing users to easily identify pollution hotspots across different regions.

Overall, the future scope of this project lies in expanding its data coverage, enhancing its analytical techniques, and integrating advanced technologies to create a more robust and intelligent air quality monitoring system. Such improvements would contribute to better environmental management and support informed decision-making for reducing air pollution.

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The preparation of this project involved referring to a variety of books, online resources, and documentation related to data science, Python programming, and air quality analysis. Standard textbooks such as *Python for Data Analysis* by Wes McKinney and *Python Data Science Handbook* by Jake VanderPlas were used to understand data manipulation, pre-processing, and visualization techniques.

In addition to books, official documentation of Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn was referred to for implementing various data analysis and visualization tasks. These resources provided detailed explanations and practical examples that were helpful during the development of the project.

Online learning platforms, tutorials, and articles were also used to gain a better understanding of exploratory data analysis (EDA), data cleaning methods, and handling real-world datasets. The dataset itself was accessed from Kaggle, which serves as a reliable platform for obtaining structured datasets for analysis. Furthermore, background knowledge about air pollution and its impact on health was supported by information available from official environmental and health organizations.

Overall, these resources played a significant role in successfully completing the project by providing both theoretical knowledge and practical guidance for implementation.

GitHub Link: https://github.com/thahseenjafarpp/air-quality-analysis

LinkedIn Link: https://www.linkedin.com/feed/update/urn:li:activity:7453114226891575296/